Problems on Propositional logic

• Convert to logic:
– It is raining, it is snowing or it is dry.
– It is warm.
– It is not raining raining.
– It is not snowing.
– If the weather is nice, then it is good to walk.
– If the weather is dry and warm, the weather is nice.
Prove by resolution: “It is good to walk

A knowledge base has the following statements:
🕐 If there is gas in the tank and the fuel line is okay, then there is gas in the engine;
🕐 If there is gas in the engine and a good spark, the engine runs;
🕐 If there is power to the plugs and the plugs are clean, a good spark is produced;
🕐 If the battery is charged and the cables are okay, then there is power to the plugs.

a. Convert the rules above to CNF using proposition symbols such as GasInTank, FuelLineOK, GasInEngine, etc.
b. Suppose that you are given the facts that there is gas in the tank, the battery is charged, the fuel line and cables are both okay, and the plugs are clean. Using resolution, prove that the engine runs